

BPCL RO NETWORK PLATFORM

SYSTEM PERFORMANCE & BENEFITS REPORT

*A Comparative Study of Automated Network Ingestion
vs. Manual Excel Spreadsheet Calculations*

Executive Summary & System Evaluation

This document provides a comprehensive breakdown of the Bharat Petroleum Corporation Limited Retail Outlet (RO) Monitoring Platform. It evaluates the capabilities of the automated, web-based analytics dashboard and details the strategic advantages of transitioning from manual spreadsheet methods to automated telemetry calculations.

ORGANIZATION:

Bharat Petroleum Corporation Limited (BPCL)
Retail Business Unit, RO Systems Performance

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1. Executive Summary & Core Challenge

Evaluating spreadsheet limits in BPCL Retail Outlet telemetry analytics

Bharat Petroleum Corporation Limited manages a massive network of retail outlets equipped with Multi-Product Dispensers (MPD) and Automatic Tank Gauges (ATG). Ingesting daily logs, auditing device heartbeats, and computing uptime ratios is critical for ensuring continuous retail operations and managing vendor service level agreements (SLAs).

The Traditional Workflow: Manual Excel Spreadsheets

Historically, administrators gathered daily device logs via CSV/Excel exports from various vendors and compiled them manually. While spreadsheets are convenient for initial data entry, they face severe operational limitations when used as an analytics engine for large-scale operations:

- * **Formula Fragility:** Excel cells depend on relative coordinates. A single deleted row or altered column position completely breaks VLOOKUP, SUMIFS, and AVERAGE formulas, leading to silent calculation errors.
- * **Scalability Boundaries:** As the fleet grows to hundreds of sites, spreadsheets become sluggish, frequently crash, and are highly difficult to share securely across multiple departments.
- * **No Single Source of Truth:** Multiple copies of spreadsheets ('v1', 'v2_edited', 'v3_final') circulate via emails, resulting in conflicting datasets, mismatched numbers, and an inability to track historical trends.
- * **Manual Gap Analysis:** Calculating differences between IoT-enabled and Non-IoT sites requires writing complex macros, pivot tables, and custom scripts, creating a heavy burden on administrative staff every week.

The Solution: BPCL RO Network Performance Platform

The RO Performance Platform replaces this manual overhead with a robust, pre-compiled web application backed by an automated relational database. By standardizing the business logic inside the API server, the system allows BPCL operators to upload Excel sheets directly, instantly recalculating the entire network's analytics, gaps, and vendor scorecards in under a second.

2. Platform Modules & Features

Detailed explanation of the web application pages and capabilities

1. Overview Dashboard

The centralized command center that visualizes the current health of the entire RO network. It features live KPI summary tiles showing Total ROs, Online counts, IoT coverage, and dynamic percentage breakdowns. It hosts symmetrical donut charts for both the MPD Network and Tank (ATG) Network, letting operators visualize network status at a single glance. Below the charts, an interactive 'Network Table' tab shows the full list of ROs.

2. Sites Directory

A highly searchable tabular index of all onboarded retail outlets. Unlike a spreadsheet where filtering requires creating custom views, the Sites page contains custom inline dropdown filters directly inside the column headers. Operators can filter instantly by Sales Area, Vendor Name, MPD Offline status, and performance thresholds (e.g. isolating sites where MPD Uptime is below 70%).

3. Sales Wise & Vendor Wise RO Performance

Dedicated intelligence modules that group data dynamically by geographic area or supplier. These pages compile absolute device counts (Boarded, Online, and Offline) and render them in vertical bar charts with steeper rotated labels for readability. Below the charts, the platform lists a full interactive scorecard. Users can click any Sales Area or Vendor row to instantly drill down into a list of specific ROs belonging to that group.

4. IoT Enabled vs. Non-IoT Analysis

A specialized module designed to track the performance benefits of IoT deployments. The page calculates the MPD Online Gap and Tank Online Gap, demonstrating exactly how much better IoT-enabled sites perform compared to Non-IoT sites. It hosts a full comparison matrix by Sales Area, listing total RO counts, status distributions, MPD online ratios, Tank online ratios, and the direct percentage gap between the two groups.

5. Upload Data Center

The ingestion gateway. Operators simply drag and drop vendor Excel spreadsheets. The backend automatically parses the worksheets, maps the columns, validates the telemetry values, updates the database, and immediately refreshes all metrics across the platform.

3. Excel vs. RO Intelligence Platform

Direct feature comparison showing operational benefits

The following matrix outlines the technical and operational differences between performing calculations manually in Microsoft Excel versus using the automated RO Intelligence Platform:

Feature	Manual Excel Process	RO Intelligence Platform
Data Entry	Manual copy-pasting from emails & logs	One-click drag-and-drop Excel file upload
Human Error Risk	High (formula breaks, deleted rows)	Zero (server-validated business logic)
Processing Speed	15-30 minutes per log spreadsheet	Instantaneous (< 1 second) processing
Cross-Filtering	Difficult to align across multiple sheets	Custom column-heading filters & dropdowns
Symmetrical Tracking	Requires separate formulas for MPD vs Tank	Nested database records tracked together
Historical Trends	Hard to compile without slowing workbook	Maintained in DB for timeline trends
Drill-Down Analytics	Manual sheet filtering or pivot updates	Interactive click-to-drill down by area/vendor
Vendor Performance	Difficult to group and rank vendors	Live SLA charts & vendor comparison tables

Operational Uptime Impact

By moving calculations to the web database, BPCL reduces its analytical overhead from hours to seconds. More importantly, it provides administrative staff with clean, structured, and filterable telemetry data that allows them to identify and resolve device outages immediately, directly translating to higher retail sales.

4. Business Value & Recommendations

Quantifying the return on using the RO Intelligence Platform

Key Value Drivers

- * **95% Time Efficiency Improvement:** Manual spreadsheet compilation that previously took hours of weekly effort is now handled instantly upon data upload, freeing up regional managers to focus on maintenance SLAs.
- * **Proactive Offline Hotspot Detection:** Instead of discovering outages at the end of the week, the 'Offline Hotspots' tab automatically highlights faulty MPDs and Tanks immediately, enabling prompt maintenance dispatches.
- * **Objective Vendor SLA Auditing:** The platform groups MPD and Tank online ratios by Vendor (e.g. PINELABS, EVIDEN, GEMS, BCT). This gives BPCL procurement officers unalterable, data-backed reports to hold vendors accountable to their uptime contracts.
- * **Centralized Security & Collaboration:** The platform runs on a local/cloud dev server. Anyone with access can view the exact same statistics simultaneously, avoiding spreadsheet version-control conflicts.

Technical Architecture Highlights

The system is built using modern, lightweight, and scalable technologies designed for maximum responsiveness:

- * **FastAPI Backend:** Asynchronous Python web server that performs rapid database queries, aggregates telemetry counts, and handles file uploads securely.
- * **React Frontend:** Responsive UI dashboard using clean styled components and Recharts. Ticks are rotated to -45 degrees with reserved space to ensure charts are readable at any resolution.
- * **Centralized Relational Database:** Stores site details, vendors, sales areas, and historical sensor readings, preserving a clean history for future trend forecasting.

Recommendation:

BPCL managers should deprecate local Excel trackers for daily reporting and enforce the RO Intelligence Platform as the single system of record. Weekly regional meetings should utilize the live 'IoT Enabled vs Non-IoT' analysis to drive network upgrade decisions.